

Indian Statistical Institute
M. Math 2nd year
Academic year 2025-2026
Fourth semester
Endsem Examination
Course: Teichmüller theory
27 - 04 - 26
3 hours

- Answer as many questions as you can.
- You may use results proved in class, but make sure to state them clearly.
- Maximum marks is 60.
- Unless otherwise mentioned, all Riemann surfaces are assumed to be connected.

1. Let $X_i = \mathbb{H}/\Gamma_i, i = 1, 2$ be Riemann surfaces, let $f : X_1 \rightarrow X_2$ be a QC homeomorphism, let $F : \mathbb{H} \rightarrow \mathbb{H}$ be a lift of f , and let $\mu = \mu_F$ be the Beltrami coefficient of F . Show that for all $\gamma \in \Gamma_1$, $\mu(z) = \mu(\gamma(z)) \cdot \overline{\gamma'(z)}/\gamma'(z)$ for a.e. $z \in \mathbb{H}$. (8 marks)
2. Let L be a lattice in $\mathbb{C}$, let $X = \mathbb{C}/L$, let $\pi : \mathbb{C} \rightarrow X$ be the universal covering map, and let $\{z_\alpha\}_{\alpha \in I}$ be the collection of charts on X given by local inverses of π . For $c \in \mathbb{C}$ let $T_c \in \text{Aut}(X)$ be the map $T_c : X \rightarrow X, z + L \mapsto z + c + L$. By a line segment in X we mean a curve $\gamma : I \subset \mathbb{R} \rightarrow X$ in X such that any lift of γ is a straight line segment in $\mathbb{C}$.
- a) For $f \in \text{Aut}(X)$, show that if f is homotopic to the identity then $f = T_c$ for some $c \in \mathbb{C}$.
- b) Let $g : X \rightarrow X$ be a QC map which is homotopic to the identity. Show that there exists a constant $M > 0$ such that $l(g(\gamma)) \geq l(\gamma) - 2M$ for all line segments γ in X (here $l(\gamma)$ denotes the length of a curve γ with respect to the canonical flat metric on X).

Let $g : X \rightarrow X$ be a QC map which is homotopic to the identity. Let λ_g be the measurable function on X defined by $\lambda_g(p) = |\frac{\partial}{\partial x}(z_\beta \circ g \circ z_\alpha^{-1})(z_\alpha(p))|$ for a.e. $p \in X$, where z_α, z_β are charts near p and $g(p)$ respectively. Show that

$$\frac{1}{\text{Area}(X)} \int_X \lambda_g \, d\text{vol}_X \geq 1$$

(here $d\text{vol}_X$ and $\text{Area}(X)$ denote the area form on X and the area of X with respect to the canonical flat metric on X). (6+8+8 = 22 marks)

3. Let X, Y be compact Riemann surfaces of genus $g \geq 2$ and let $f : X \rightarrow Y$ be a QC map. Show that there exists a QC map $f_0 : X \rightarrow Y$ such that f_0 is homotopic to f and such that $K(f_0) \leq K(h)$ for all QC maps $h : X \rightarrow Y$ which are homotopic to f (10 marks)
4. a) Show that the mapping class group Mod_g acts by isometries on the Teichmüller space $\mathcal{T}_g$.
b) Let (X, f) be a marked Riemann surface of genus $g \geq 2$. Show that the stabilizer of $[X, f] \in \mathcal{T}_g$ in the mapping class group Mod_g is isomorphic to $\text{Aut}(X)$. (6+8 = 14 marks)
5. Let $g \geq 2$ and let $[X, h] \in \mathcal{T}_g$. Let $\phi \in Q(X) - \{0\}$, let $a_t \cdot (X, \phi) = (X_t, \phi_t)$ for $t \in \mathbb{R}$ where $a_t \in \text{SL}(2, \mathbb{R})$ is the diagonal matrix with diagonal entries $e^{t/2}, e^{-t/2}$, and let $f_t : X \rightarrow X_t$ be the corresponding Teichmüller mapping with initial and terminal differentials ϕ and ϕ_t respectively such that $K(f_t) = e^t$.
a) Show that the curve $\gamma_0(t) = [X_t, f_t \circ h] \in \mathcal{T}_g$ is a geodesic in $\mathcal{T}_g$.
b) Show that if $\gamma : [0, T] \rightarrow \mathcal{T}_g$ is a geodesic such that $\gamma(0) = \gamma_0(0), \gamma(T) = \gamma_0(T)$, then $\gamma(t) = \gamma_0(t)$ for all $t \in [0, T]$. (8+8 = 16 marks)